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APM1220 - Applied Multivariate Data Analysis

INSTRUCTIONS:

- Answer all questions completely and show all necessary steps, intermediate calculations, and mathematical proofs clearly to receive full credit.
- Where matrix operations are required, ensure that dimensions conform before performing any multiplications, additions, or inversions.
- Present all final matrix entries and numerical results in exact, simplified form (e.g., fractions or radicals) unless decimals are specifically requested.
- Clearly label your final answers for each sub-problem (a, b, c, etc.).

Problem 2.2 (5 points)

Given the matrices

$$A = \begin{bmatrix} -1 & 3 \\ 4 & 2 \end{bmatrix}, \quad B = \begin{bmatrix} 4 & -3 \\ 1 & -2 \\ -2 & 0 \end{bmatrix}, \quad \text{and} \quad C = \begin{bmatrix} 5 \\ -4 \\ 2 \end{bmatrix}$$

perform the indicated multiplications:

(a) $5A$

Sol.

$$\begin{aligned} 5A &= 5 \begin{bmatrix} -1 & 3 \\ 4 & 2 \end{bmatrix} \\ &= \begin{bmatrix} 5(-1) & 5(3) \\ 5(4) & 5(2) \end{bmatrix} \\ &= \begin{bmatrix} -5 & 15 \\ 20 & 10 \end{bmatrix} \end{aligned}$$

Hence,

$$\boxed{5A = \begin{bmatrix} -5 & 15 \\ 20 & 10 \end{bmatrix}}$$

(b) BA

Sol.

$$\begin{aligned} BA &= \begin{bmatrix} 4 & -3 \\ 1 & -2 \\ -2 & 0 \end{bmatrix} \begin{bmatrix} -1 & 3 \\ 4 & 2 \end{bmatrix} \\ BA &= \begin{bmatrix} 4(-1) + (-3)(4) & 4(3) + (-3)(2) \\ 1(-1) + (-2)(4) & 1(3) + (-2)(2) \\ (-2)(-1) + 0(4) & (-2)(3) + 0(2) \end{bmatrix} \\ &= \begin{bmatrix} -4 - 12 & 12 - 6 \\ -1 - 8 & 3 - 4 \\ 2 + 0 & -6 + 0 \end{bmatrix} \\ &= \begin{bmatrix} -16 & 6 \\ -9 & -1 \\ 2 & -6 \end{bmatrix} \end{aligned}$$

Hence,

$$BA = \begin{bmatrix} -16 & 6 \\ -9 & -1 \\ 2 & -6 \end{bmatrix}$$

(c) $A'B'$

Sol.

The transpose of A is

$$A' = \begin{bmatrix} -1 & 4 \\ 3 & 2 \end{bmatrix}$$

and the transpose of B is

$$B' = \begin{bmatrix} 4 & 1 & -2 \\ -3 & -2 & 0 \end{bmatrix}.$$

So,

$$A'B' = \begin{bmatrix} -1 & 4 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} 4 & 1 & -2 \\ -3 & -2 & 0 \end{bmatrix}$$

$$\begin{aligned}
&= \begin{bmatrix} (-1)(4) + 4(-3) & (-1)(1) + 4(-2) & (-1)(-2) + 4(0) \\ 3(4) + 2(-3) & 3(1) + 2(-2) & 3(-2) + 2(0) \end{bmatrix} \\
&= \begin{bmatrix} -4 - 12 & -1 - 8 & 2 + 0 \\ 12 - 6 & 3 - 4 & -6 + 0 \end{bmatrix} \\
&= \begin{bmatrix} -16 & -9 & 2 \\ 6 & -1 & -6 \end{bmatrix}
\end{aligned}$$

Hence,

$$\boxed{A'B' = \begin{bmatrix} -16 & -9 & 2 \\ 6 & -1 & -6 \end{bmatrix}}$$

(d) $C'B$

Sol.

The transpose of C is

$$C' = \begin{bmatrix} 5 & -4 & 2 \end{bmatrix}.$$

So,

$$\begin{aligned}
C'B &= \begin{bmatrix} 5 & -4 & 2 \end{bmatrix} \begin{bmatrix} 4 & -3 \\ 1 & -2 \\ -2 & 0 \end{bmatrix} \\
&= \begin{bmatrix} 5(4) + (-4)(1) + 2(-2) & 5(-3) + (-4)(-2) + 2(0) \end{bmatrix} \\
&= \begin{bmatrix} 20 - 4 - 4 & -15 + 8 + 0 \end{bmatrix} \\
&= \begin{bmatrix} 12 & -7 \end{bmatrix}
\end{aligned}$$

Hence,

$$\boxed{C'B = \begin{bmatrix} 12 & -7 \end{bmatrix}}$$

(e) Is AB defined?

Sol.

The dimensions of the matrices are

$$A_{2 \times 2} \quad \text{and} \quad B_{3 \times 2}.$$

For the multiplication AB to be defined, the number of columns of A must equal the number of rows of B .

However,

$$2 \neq 3.$$

Therefore, the dimensions do not conform for matrix multiplication.
Hence,

AB is not defined.

Problem 2.4 (6 points)

When A^{-1} and B^{-1} exist, prove each of the following:

(a) $(A')^{-1} = (A^{-1})'$

Sol.

Since A^{-1} exists,

$$AA^{-1} = I$$

Transpose both sides

$$(AA^{-1})' = I'$$

Using the property

$$(AB)' = B'A'$$

we have

$$(A^{-1})'A' = I'$$

Since the transpose of the identity matrix is itself,

$$I' = I$$

Thus,

$$(A^{-1})'A' = I$$

Since

$$A^{-1}A = I$$

taking the transpose gives

$$(A^{-1}A)' = I'$$

Therefore,

$$A'(A^{-1})' = I$$

Hence, $(A^{-1})'$ is the inverse of A' and,

$$\boxed{(A')^{-1} = (A^{-1})'}$$

(b) $(AB)^{-1} = B^{-1}A^{-1}$

Sol.

Consider the product

$$(B^{-1}A^{-1})AB$$

Using the associative property of matrix multiplication

$$(B^{-1}A^{-1})AB = B^{-1}(A^{-1}A)B$$

Since

$$A^{-1}A = I$$

we have

$$B^{-1}(A^{-1}A)B = B^{-1}IB$$

Since multiplication by the identity matrix does not change a matrix

$$B^{-1}IB = B^{-1}B$$

Since

$$B^{-1}B = I$$

we obtain

$$(B^{-1}A^{-1})AB = I$$

Now consider the product in the opposite order

$$AB(B^{-1}A^{-1})$$

Using the associative property

$$AB(B^{-1}A^{-1}) = A(BB^{-1})A^{-1}$$

Since

$$BB^{-1} = I$$

we have

$$A(BB^{-1})A^{-1} = AIA^{-1}$$

Since multiplication by the identity matrix does not change a matrix

$$AIA^{-1} = AA^{-1}$$

Since

$$AA^{-1} = I$$

we obtain

$$AB(B^{-1}A^{-1}) = I$$

Therefore $B^{-1}A^{-1}$ is the inverse of AB

Hence

$$\boxed{(AB)^{-1} = B^{-1}A^{-1}}$$

Problem 2.8 (6 points)

Given the matrix

$$A = \begin{bmatrix} 1 & 2 \\ 2 & -2 \end{bmatrix}$$

find the eigenvalues λ_1 and λ_2 and the associated normalized eigenvectors e_1 and e_2
Determine the spectral decomposition (2-16) of A

Sol.

To find the eigenvalues solve the characteristic equation

$$|A - \lambda I| = 0$$

$$\begin{vmatrix} 1 - \lambda & 2 \\ 2 & -2 - \lambda \end{vmatrix} = 0$$

$$(1 - \lambda)(-2 - \lambda) - 4 = 0$$

$$-2 - \lambda + 2\lambda + \lambda^2 - 4 = 0$$

$$\lambda^2 + \lambda - 6 = 0$$

$$(\lambda - 2)(\lambda + 3) = 0$$

Hence,

$$\boxed{\lambda_1 = 2 \quad \lambda_2 = -3}$$

Finding the eigenvector associated with $\lambda_1 = 2$

Solve

$$(A - 2I)e_1 = 0$$

$$\begin{aligned} A - 2I &= \begin{bmatrix} 1 - 2 & 2 \\ 2 & -2 - 2 \end{bmatrix} \\ &= \begin{bmatrix} -1 & 2 \\ 2 & -4 \end{bmatrix} \end{aligned}$$

Let

$$e_1 = \begin{bmatrix} x \\ y \end{bmatrix}$$

Then

$$\begin{bmatrix} -1 & 2 \\ 2 & -4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Using the first equation

$$-x + 2y = 0$$

$$x = 2y$$

Let

$$y = 1$$

Then

$$x = 2$$

An eigenvector corresponding to $\lambda_1 = 2$ is

$$\begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

The length of the eigenvector is

$$\begin{aligned} &\sqrt{2^2 + 1^2} \\ &= \sqrt{5} \end{aligned}$$

Hence,

$$e_1 = \frac{1}{\sqrt{5}} \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

Finding the eigenvector associated with $\lambda_2 = -3$

Solve

$$(A + 3I)e_2 = 0$$

$$\begin{aligned} A + 3I &= \begin{bmatrix} 1+3 & 2 \\ 2 & -2+3 \end{bmatrix} \\ &= \begin{bmatrix} 4 & 2 \\ 2 & 1 \end{bmatrix} \end{aligned}$$

Let

$$e_2 = \begin{bmatrix} x \\ y \end{bmatrix}$$

Then

$$\begin{bmatrix} 4 & 2 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Using the first equation

$$4x + 2y = 0$$

$$2x + y = 0$$

$$y = -2x$$

Let

$$x = 1$$

Then

$$y = -2$$

An eigenvector corresponding to $\lambda_2 = -3$ is

$$\begin{bmatrix} 1 \\ -2 \end{bmatrix}$$

The length of the eigenvector is

$$\sqrt{1^2 + (-2)^2}$$

$$= \sqrt{5}$$

Hence,

$$e_2 = \frac{1}{\sqrt{5}} \begin{bmatrix} 1 \\ -2 \end{bmatrix}$$

Finding the spectral decomposition

Since A is symmetric its spectral decomposition is

$$A = \lambda_1 e_1 e_1' + \lambda_2 e_2 e_2'$$

Substituting the eigenvalues and normalized eigenvectors

$$A = 2e_1 e_1' - 3e_2 e_2'$$

First calculate $e_1 e_1'$

$$\begin{aligned} e_1 e_1' &= \frac{1}{\sqrt{5}} \begin{bmatrix} 2 \\ 1 \end{bmatrix} \frac{1}{\sqrt{5}} \begin{bmatrix} 2 & 1 \end{bmatrix} \\ &= \frac{1}{5} \begin{bmatrix} 2 \\ 1 \end{bmatrix} \begin{bmatrix} 2 & 1 \end{bmatrix} \\ &= \frac{1}{5} \begin{bmatrix} 4 & 2 \\ 2 & 1 \end{bmatrix} \end{aligned}$$

Now calculate $e_2 e'_2$

$$\begin{aligned} e_2 e'_2 &= \frac{1}{\sqrt{5}} \begin{bmatrix} 1 \\ -2 \end{bmatrix} \frac{1}{\sqrt{5}} [1 \quad -2] \\ &= \frac{1}{5} \begin{bmatrix} 1 \\ -2 \end{bmatrix} [1 \quad -2] \\ &= \frac{1}{5} \begin{bmatrix} 1 & -2 \\ -2 & 4 \end{bmatrix} \end{aligned}$$

Therefore,

$$\begin{aligned} A &= 2 \left(\frac{1}{5} \begin{bmatrix} 4 & 2 \\ 2 & 1 \end{bmatrix} \right) - 3 \left(\frac{1}{5} \begin{bmatrix} 1 & -2 \\ -2 & 4 \end{bmatrix} \right) \\ &= \frac{1}{5} \begin{bmatrix} 8 & 4 \\ 4 & 2 \end{bmatrix} - \frac{1}{5} \begin{bmatrix} 3 & -6 \\ -6 & 12 \end{bmatrix} \\ &= \frac{1}{5} \left[\begin{bmatrix} 8 & 4 \\ 4 & 2 \end{bmatrix} - \begin{bmatrix} 3 & -6 \\ -6 & 12 \end{bmatrix} \right] \\ &= \frac{1}{5} \begin{bmatrix} 5 & 10 \\ 10 & -10 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 2 \\ 2 & -2 \end{bmatrix} \end{aligned}$$

Hence,

$$\boxed{A = 2e_1 e'_1 - 3e_2 e'_2}$$

where

$$\boxed{e_1 = \frac{1}{\sqrt{5}} \begin{bmatrix} 2 \\ 1 \end{bmatrix} \quad e_2 = \frac{1}{\sqrt{5}} \begin{bmatrix} 1 \\ -2 \end{bmatrix}}$$

Problem 2.11 (5 points)

Show that the determinant of the $p \times p$ diagonal matrix $A = \{a_{ij}\}$ with $a_{ij} = 0$ for $i \neq j$ is given by the product of the diagonal elements

$$|A| = a_{11} a_{22} \cdots a_{pp}$$

Sol.

Consider the $p \times p$ diagonal matrix

$$A = \begin{bmatrix} a_{11} & 0 & 0 & \cdots & 0 \\ 0 & a_{22} & 0 & \cdots & 0 \\ 0 & 0 & a_{33} & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & a_{pp} \end{bmatrix}$$

By the definition of the determinant expand along the first row

$$|A| = a_{11}A_{11} + 0 + \cdots + 0$$

$$|A| = a_{11}A_{11}$$

Delete the first row and first column

$$A_{11} = \begin{vmatrix} a_{22} & 0 & \cdots & 0 \\ 0 & a_{33} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_{pp} \end{vmatrix}$$

Expand along the first row

$$A_{11} = a_{22}A_{22}$$

Substitute into the determinant of A

$$|A| = a_{11}a_{22}A_{22}$$

$$|A| = a_{11}a_{22}a_{33} \cdots a_{pp}$$

Hence, the determinant of a diagonal matrix is the product of its diagonal elements

$$\boxed{|A| = a_{11}a_{22} \cdots a_{pp}}$$

Problem 2.24 (5 points)

Let X have covariance matrix

$$\Sigma = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Find:

(a) Σ^{-1}

(b) The eigenvalues and eigenvectors of Σ

(c) The eigenvalues and eigenvectors of Σ^{-1}

Answers:

(a) Σ^{-1}

Sol.

Since Σ is a diagonal matrix its inverse is obtained by taking the reciprocal of each diagonal element

Hence,

$$\Sigma^{-1} = \begin{bmatrix} \frac{1}{4} & 0 & 0 \\ 0 & \frac{1}{9} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

(b) The eigenvalues and eigenvectors of Σ

Sol.

Solve for

$$|\Sigma - \lambda I| = 0$$

$$\begin{vmatrix} 4 - \lambda & 0 & 0 \\ 0 & 9 - \lambda & 0 \\ 0 & 0 & 1 - \lambda \end{vmatrix} = 0$$

$$(4 - \lambda)(9 - \lambda)(1 - \lambda) = 0$$

Hence, the eigenvalues are

$$\lambda_1 = 4 \quad \lambda_2 = 9 \quad \lambda_3 = 1$$

Eigenvector corresponding to $\lambda_1 = 4$

Solve for

$$(\Sigma - 4I)e_1 = 0$$

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & -3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

This gives

$$5y = 0$$

$$-3z = 0$$

$$y = 0 \quad z = 0$$

Let

$$x = 1$$

Hence,

$$e_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

Eigenvector corresponding to $\lambda_2 = 9$

Solve for

$$(\Sigma - 9I)e_2 = 0$$

$$\begin{bmatrix} -5 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -8 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

This gives

$$-5x = 0$$

$$-8z = 0$$

$$x = 0 \quad z = 0$$

Let

$$y = 1$$

Hence,

$$e_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

Eigenvector corresponding to $\lambda_3 = 1$

Solve for

$$(\Sigma - I)e_3 = 0$$

$$\begin{bmatrix} 3 & 0 & 0 \\ 0 & 8 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

This gives

$$3x = 0$$

$$8y = 0$$

$$x = 0 \quad y = 0$$

Let

$$z = 1$$

Hence,

$$e_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Hence, the eigenvalues and corresponding eigenvectors of Σ are

$$\lambda_1 = 4 \quad e_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$\lambda_2 = 9 \quad e_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$$\lambda_3 = 1 \quad e_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

(c) The eigenvalues and eigenvectors of Σ^{-1}

Sol.

$$\Sigma^{-1} = \begin{bmatrix} \frac{1}{4} & 0 & 0 \\ 0 & \frac{1}{9} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Since the eigenvalues of the inverse of a matrix are the reciprocals of the nonzero eigenvalues of the original matrix

$$\lambda_1^* = \frac{1}{4}$$

$$\lambda_2^* = \frac{1}{9}$$

$$\lambda_3^* = 1$$

The eigenvectors remain unchanged because

$$\Sigma e_i = \lambda_i e_i$$

$$\Sigma^{-1} e_i = \frac{1}{\lambda_i} e_i$$

Hence,

$$\lambda_1^* = \frac{1}{4} \quad e_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$\lambda_2^* = \frac{1}{9} \quad e_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$$\lambda_3^* = 1 \quad e_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Problem 2.30 (8 points)

You are given the random vector

$$X' = [X_1, X_2, X_3, X_4]$$

with mean vector

$$\mu'_X = [4, 3, 2, 1]$$

and variance-covariance matrix

$$\Sigma_X = \begin{bmatrix} 3 & 0 & 2 & 2 \\ 0 & 1 & 1 & 0 \\ 2 & 1 & 9 & -2 \\ 2 & 0 & -2 & 4 \end{bmatrix}$$

Partition X as

$$X = \begin{bmatrix} X_1 \\ X_2 \\ X_3 \\ X_4 \end{bmatrix} = \begin{bmatrix} X^{(1)} \\ X^{(2)} \end{bmatrix}$$

Let

$$A = \begin{bmatrix} 1 & 2 \end{bmatrix}$$

and

$$B = \begin{bmatrix} 1 & -2 \\ 2 & -1 \end{bmatrix}$$

and consider the linear combinations $AX^{(1)}$ and $BX^{(2)}$

(a) $E(X^{(1)})$

Sol.

The first two elements of the mean vector form the mean vector of $X^{(1)}$

Hence,

$$E(X^{(1)}) = \begin{bmatrix} 4 \\ 3 \end{bmatrix}$$

(b) $E(AX^{(1)})$

Sol.

Using the property

$$E(AX^{(1)}) = AE(X^{(1)})$$

$$E(AX^{(1)}) = \begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} 4 \\ 3 \end{bmatrix}$$

$$= 1(4) + 2(3)$$

$$= 4 + 6$$

$$= 10$$

Hence,

$$E(AX^{(1)}) = 10$$

(c) $\text{Cov}(X^{(1)})$

Sol.

The covariance matrix of $X^{(1)}$ is the upper-left 2×2 submatrix of Σ_X

$$\text{Cov}(X^{(1)}) = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}$$

(d) $\text{Cov}(AX^{(1)})$

Sol.

Using the property

$$\text{Cov}(AX^{(1)}) = A \text{Cov}(X^{(1)})A'$$

$$\begin{aligned}
\text{Cov}(AX^{(1)}) &= \begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} \\
&= \begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix} \\
&= \begin{bmatrix} 3 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} \\
&= 3(1) + 4(2) \\
&= 3 + 8 \\
&= 11
\end{aligned}$$

Hence,

$$\boxed{\text{Cov}(AX^{(1)}) = 11}$$

(e) $E(X^{(2)})$

Sol.

The last two elements of the mean vector form the mean vector of $X^{(2)}$

Hence,

$$\boxed{E(X^{(2)}) = \begin{bmatrix} 2 \\ 1 \end{bmatrix}}$$

(f) $E(BX^{(2)})$

Sol.

Using the property

$$E(BX^{(2)}) = BE(X^{(2)})$$

$$E(BX^{(2)}) = \begin{bmatrix} 1 & -2 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$\begin{aligned}
&= \begin{bmatrix} 1(2) + (-2)(1) \\ 2(2) + (-1)(1) \end{bmatrix} \\
&= \begin{bmatrix} 2 - 2 \\ 4 - 1 \end{bmatrix} \\
&= \begin{bmatrix} 0 \\ 3 \end{bmatrix}
\end{aligned}$$

Hence,

$$E(BX^{(2)}) = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$$

(g) $\text{Cov}(X^{(2)})$

Sol.

The covariance matrix of $X^{(2)}$ is the lower-right 2×2 submatrix of Σ_X

Hence,

$$\text{Cov}(X^{(2)}) = \begin{bmatrix} 9 & -2 \\ -2 & 4 \end{bmatrix}$$

(h) $\text{Cov}(BX^{(2)})$

Sol.

Using the property

$$\text{Cov}(BX^{(2)}) = B \text{Cov}(X^{(2)}) B'$$

$$\begin{aligned}
\text{Cov}(BX^{(2)}) &= \begin{bmatrix} 1 & -2 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 9 & -2 \\ -2 & 4 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ -2 & -1 \end{bmatrix} \\
B \text{Cov}(X^{(2)}) &= \begin{bmatrix} 1(9) + (-2)(-2) & 1(-2) + (-2)(4) \\ 2(9) + (-1)(-2) & 2(-2) + (-1)(4) \end{bmatrix} \\
&= \begin{bmatrix} 9 + 4 & -2 - 8 \\ 18 + 2 & -4 - 4 \end{bmatrix} \\
&= \begin{bmatrix} 13 & -10 \\ 20 & -8 \end{bmatrix}
\end{aligned}$$

Therefore,

$$\begin{aligned}
\text{Cov}(BX^{(2)}) &= \begin{bmatrix} 13 & -10 \\ 20 & -8 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ -2 & -1 \end{bmatrix} \\
&= \begin{bmatrix} 13(1) + (-10)(-2) & 13(2) + (-10)(-1) \\ 20(1) + (-8)(-2) & 20(2) + (-8)(-1) \end{bmatrix} \\
&= \begin{bmatrix} 13 + 20 & 26 + 10 \\ 20 + 16 & 40 + 8 \end{bmatrix} \\
&= \begin{bmatrix} 33 & 36 \\ 36 & 48 \end{bmatrix}
\end{aligned}$$

Hence,

$$\boxed{\text{Cov}(BX^{(2)}) = \begin{bmatrix} 33 & 36 \\ 36 & 48 \end{bmatrix}}$$

(i) $\text{Cov}(X^{(1)}, X^{(2)})$

Sol.

The covariance matrix between $X^{(1)}$ and $X^{(2)}$ is the upper-right 2×2 block of Σ_X

Hence,

$$\boxed{\text{Cov}(X^{(1)}, X^{(2)}) = \begin{bmatrix} 2 & 2 \\ 1 & 0 \end{bmatrix}}$$

(j) $\text{Cov}(AX^{(1)}, BX^{(2)})$

Sol.

Using the covariance transformation property

$$\begin{aligned}
\text{Cov}(AX^{(1)}, BX^{(2)}) &= A \text{Cov}(X^{(1)}, X^{(2)}) B' \\
&= \begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} 2 & 2 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ -2 & -1 \end{bmatrix} \\
A \text{Cov}(X^{(1)}, X^{(2)}) &= \begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} 2 & 2 \\ 1 & 0 \end{bmatrix}
\end{aligned}$$

$$= \begin{bmatrix} 1(2) + 2(1) & 1(2) + 2(0) \end{bmatrix}$$

$$= \begin{bmatrix} 4 & 2 \end{bmatrix}$$

Therefore,

$$\text{Cov}(AX^{(1)}, BX^{(2)}) = \begin{bmatrix} 4 & 2 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ -2 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} 4(1) + 2(-2) & 4(2) + 2(-1) \end{bmatrix}$$

$$= \begin{bmatrix} 4 - 4 & 8 - 2 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 6 \end{bmatrix}$$

Hence,

$$\boxed{\text{Cov}(AX^{(1)}, BX^{(2)}) = \begin{bmatrix} 0 & 6 \end{bmatrix}}$$

Problem 2.41 (5 points)

You are given the random vector

$$X' = [X_1, X_2, X_3, X_4]$$

with mean vector

$$\mu'_X = [3, 2, -2, 0]$$

and variance-covariance matrix

$$\Sigma_X = \begin{bmatrix} 3 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix}$$

Let

$$A = \begin{bmatrix} 1 & -1 & 0 & 0 \\ 1 & 1 & -2 & 0 \\ 1 & 1 & 1 & -3 \end{bmatrix}$$

(a) Find $E(AX)$, the mean of AX

Sol.

Using the property

$$E(AX) = AE(X)$$

$$\begin{aligned} E(AX) &= \begin{bmatrix} 1 & -1 & 0 & 0 \\ 1 & 1 & -2 & 0 \\ 1 & 1 & 1 & -3 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \\ -2 \\ 0 \end{bmatrix} \\ &= \begin{bmatrix} 1(3) + (-1)(2) + 0(-2) + 0(0) \\ 1(3) + 1(2) + (-2)(-2) + 0(0) \\ 1(3) + 1(2) + 1(-2) + (-3)(0) \end{bmatrix} \\ &= \begin{bmatrix} 3 - 2 \\ 3 + 2 + 4 \\ 3 + 2 - 2 \end{bmatrix} \\ &= \begin{bmatrix} 1 \\ 9 \\ 3 \end{bmatrix} \end{aligned}$$

Hence,

$$\boxed{E(AX) = \begin{bmatrix} 1 \\ 9 \\ 3 \end{bmatrix}}$$

(b) Find $\text{Cov}(AX)$, the variances and covariances of AX

Sol.

Using the covariance transformation property

$$\text{Cov}(AX) = A\Sigma_X A'$$

Since

$$\Sigma_X = 3I$$

we have

$$\text{Cov}(AX) = 3AA'$$

First calculate AA'

$$AA' = \begin{bmatrix} 1 & -1 & 0 & 0 \\ 1 & 1 & -2 & 0 \\ 1 & 1 & 1 & -3 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ -1 & 1 & 1 \\ 0 & -2 & 1 \\ 0 & 0 & -3 \end{bmatrix}$$

$$= \begin{bmatrix} 1(1) + (-1)(-1) + 0(0) + 0(0) & 1(1) + (-1)(1) + 0(-2) + 0(0) & 1(1) + (-1)(1) + 0(1) + 0(-3) \\ 1(1) + 1(-1) + (-2)(0) + 0(0) & 1(1) + 1(1) + (-2)(-2) + 0(0) & 1(1) + 1(1) + (-2)(1) + 0(-3) \\ 1(1) + 1(-1) + 1(0) + (-3)(0) & 1(1) + 1(1) + 1(-2) + (-3)(0) & 1(1) + 1(1) + 1(1) + (-3)(-3) \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 12 \end{bmatrix}$$

Hence,

$$\text{Cov}(AX) = 3 \begin{bmatrix} 2 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 12 \end{bmatrix}$$

$$= \begin{bmatrix} 6 & 0 & 0 \\ 0 & 18 & 0 \\ 0 & 0 & 36 \end{bmatrix}$$

Therefore

$$\boxed{\text{Cov}(AX) = \begin{bmatrix} 6 & 0 & 0 \\ 0 & 18 & 0 \\ 0 & 0 & 36 \end{bmatrix}}$$

The variances of the three linear combinations are therefore

$$\boxed{\text{Var}(AX)_1 = 6}$$

$$\boxed{\text{Var}(AX)_2 = 18}$$

$$\boxed{\text{Var}(AX)_3 = 36}$$

and all covariances between distinct linear combinations are zero

$$\boxed{\text{Cov}(AX_i, AX_j) = 0 \quad i \neq j}$$

(c) Which pairs of linear combinations have zero covariances?

Sol.

Let the three linear combinations be

$$Y_1 = X_1 - X_2$$

$$Y_2 = X_1 + X_2 - 2X_3$$

$$Y_3 = X_1 + X_2 + X_3 - 3X_4$$

From the covariance matrix obtained in part (b)

$$\text{Cov}(AX) = \begin{bmatrix} 6 & 0 & 0 \\ 0 & 18 & 0 \\ 0 & 0 & 36 \end{bmatrix}$$

we have

$$\text{Cov}(Y_1, Y_2) = 0$$

$$\text{Cov}(Y_1, Y_3) = 0$$

$$\text{Cov}(Y_2, Y_3) = 0$$

Therefore all pairs of linear combinations have zero covariance

(Y_1, Y_2)	(Y_1, Y_3)	(Y_2, Y_3)
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